

An updated checklist of mosquitoes (Diptera: Culicidae) from Chandigarh, India and its surrounding area.

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The affirmation about the prevalence of mosquito species at a particular place and time is very significant, not only to predict the danger of diseases or future outbreaks but also to control the vectors in time. Despite mosquitoes being medically important, the information about its faunal diversity is very scanty as far as Chandigarh in India and its nearby areas are concerned. So, this study was carried out to survey the mosquito fauna from areas in and around Chandigarh in northern India. Detailed mosquito surveys were carried out to explore the mosquito fauna from various habitats of developed urban areas, gardens, slums and surrounding villages of Chandigarh from June 2017-November 2019 using hand nets and oral aspirators. A total of 34 mosquito species belonging to 8 genera viz; Anopheles, Aedes, Armigeres, Culex, Coquillettia, Mansonia, Mimomyia and Verrallina were recorded, identified and preserved along with detailed collection data, of which eight are new records from Chandigarh. The present checklist of mosquito fauna comprising 34 species provides information on the occurrence of mosquito vectors in Chandigarh and its adjoining areas which will be beneficial for the health authorities to adopt appropriate measures in time for the control of these vectors.

Risk estimation for air travel-induced malaria transmission in central Europe - A mathematical modelling study.

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Aim of our study was to identify conditions under which malaria transmission caused by imported infectious mosquitoes or travellers could occur at large central European airports, and if such transmission could be sustained by indigenous mosquitoes. We developed a deterministic and a stochastic compartmental Susceptible-Exposed-Infectious-Recovered-Susceptible (humans)/Susceptible-Exposed-Infectious (mosquitoes) model with two mosquito (imported *Anopheles gambiae*, indigenous *A. plumbeus*) and three human (travellers, airport personnel exposed/not exposed to imported *A. gambiae*) populations. We assessed various scenarios to identify combinations of model parameters leading to ongoing malaria transmission at the airport. The number of infected airport personnel was low (five infected employees/six months) under assumptions reflecting possible future climatic conditions, current passenger mobility and no desinsection of airports/aircraft. Almost all infections among airport personnel were directly due to bites by imported *A. gambiae*. Indigenous mosquitoes would need to have comparable transmission parameters to *A. gambiae* to sustain disease transmission. Incoming infectious passengers play only a minor role in malaria transmission. Use of aircraft/airport desinsection led to no transmission events in the model. Our study shows that sustainable air travel-induced malaria transmission in central Europe is unlikely under current conditions or conditions which might become realistic in the next century.

Salivary AsHPX12 influences pre-blood meal associated behavioral properties in *Anopheles stephensi*.

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A successful blood meal acquisition process by an adult female mosquito is accomplished through salivary glands, which releases a cocktail of proteins to counteract the vertebrate host's immune homeostasis. Here, we characterize a salivary-specific Heme peroxidase family member HPX12, originally identified from *Plasmodium vivax* infected salivary RNAseq data of the mosquito *Anopheles stephensi*. To demonstrate we utilized a comprehensive in silico and functional genomics approach. Our dsRNA-mediated silencing experiments demonstrate that salivary AsHPX12 may regulate pre-blood meal-associated behavioral properties such as probing time, probing propensity, and host attraction. Altered expression of the salivary secretory and antennal proteins expression may have accounted for salivary homeostasis disruption resulting in the unusual fast release of salivary cocktail proteins and delayed acquisition of blood meal in the AsHPX12 knockdown mosquitoes. We also observed a significant parallel transcriptional modulation in response to blood feeding and *P. vivax* infection. With this work, we establish a possible functional correlation of AsHPX12 role in the maintenance of salivary physiological-homeostasis, and *Plasmodium* sporozoites survival/transmission, though the mechanism is yet to unravel.

An *Anopheles* by Any Other Name ...?

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The physical characteristics of a nematoceran fly that evokes the image of a mosquito of the genus *Anopheles* Meigen (Diptera: Culicidae) has existed since the genus was defined in the early part of the twentieth century. Is that image likely to change with the recent proposal to elevate four Neotropical subgenera to generic status based on relationships generated by phylogenetic analyses of mitochondrial protein-coding genes? Reasons for retaining the traditional concept of *Anopheles*, with its traditionally recognized subgenera, are presented. However, as the ranking of taxa as genera or subgenera is subjective, the choice of rank of a genus-group name may be considered a subjective decision and a matter of user preference.

Identification keys to the *Anopheles* mosquitoes of South America (Diptera: Culicidae). I. Introduction.

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The worldwide genus *Anopheles* Meigen, 1918 is the only genus containing species evolved as vectors of human and simian malaria. Morbidity and mortality caused by *Plasmodium* Marchiafava & Celli, 1885 is tremendous, which has made these parasites and their vectors the objects of intense research aimed at mosquito identification, malaria control and elimination. DNA tools make the identification of *Anopheles* species both easier and more difficult. Easier in that putative species can nearly always be separated based on DNA data; more difficult in that attaching a scientific name to a species is often problematic because morphological characters are often difficult to interpret or even see; and DNA technology might not be available and affordable. Added to this are the many species that are either not yet recognized or are similar to, or identical with, named species. The first step in solving *Anopheles* identification problem is to attach a morphology-based formal or informal name to a specimen. These names are hypotheses to be tested with further morphological observations and/or DNA evidence. The overarching objective is to be able to communicate about a given species under study. In South America, morphological

identification which is the first step in the above process is often difficult because of lack of taxonomic expertise and/or inadequate identification keys, written for local fauna, containing the most consequential species, or obviously, do not include species described subsequent to key publication. Holotypes and paratypes and other specimens deposited in the Coleção Entomológica de Referência, Faculdade de Saúde Pública (FSP-USP), Museo de Entomología, Universidad del Valle (MUSENUV) and the US National Mosquito Collection, Smithsonian Institution (USNMC) were examined and employed to illustrate the identification keys for female, male and fourth-instar larvae of *Anopheles*. We presented, in four concurrent parts, introduction and three keys to aid the identification of South American *Anopheles* based on the morphology of the larvae, male genitalia and adult females, with the former two keys fully illustrated. Taxonomic information and identification keys for species of the genus *Anopheles* are updated. The need for further morphology-based studies and description of new species are reinforced.

Chromatin Structure and Function in Mosquitoes.

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The principles and function of chromatin and nuclear architecture have been extensively studied in model organisms, such as *Drosophila melanogaster*. However, little is known about the role of these epigenetic processes in transcriptional regulation in other insects including mosquitoes, which are major disease vectors and a worldwide threat for human health. Some of these life-threatening diseases are malaria, which is caused by protozoan parasites of the genus *Plasmodium* and transmitted by *Anopheles* mosquitoes; dengue fever, which is caused by an arbovirus mainly transmitted by *Aedes aegypti*; and West Nile fever, which is caused by an arbovirus transmitted by *Culex* spp. In this contribution, we review what is known about chromatin-associated mechanisms and the 3D genome structure in various mosquito vectors, including *Anopheles*, *Aedes*, and *Culex* spp. We also discuss the similarities between epigenetic mechanisms in mosquitoes and the model organism *Drosophila melanogaster*, and advocate that the field could benefit from the cross-application of state-of-the-art functional genomic technologies that are well-developed in the fruit fly. Uncovering the mosquito regulatory genome can lead to the discovery of unique regulatory networks associated with the parasitic life-style of these insects. It is also critical to understand the molecular interactions between the vectors and the pathogens that they transmit, which could hold the key to major breakthroughs on the fight against mosquito-borne diseases. Finally, it is clear that epigenetic mechanisms controlling mosquito environmental plasticity and evolvability are also of utmost importance, particularly in the current context of globalization and climate change.

A targeted amplicon sequencing panel to simultaneously identify mosquito species and *Plasmodium* presence across the entire *Anopheles* genus.

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Anopheles is a diverse genus of mosquitoes comprising over 500 described species, including all known human malaria vectors. While a limited number of key vector species have been studied in detail, the goal of malaria elimination calls for surveillance of all potential vector species. Here, we develop a multilocus amplicon sequencing approach that targets 62 highly variable loci in the *Anopheles* genome and two conserved loci in the *Plasmodium* mitochondrion, simultaneously revealing both the mosquito

species and whether that mosquito carries malaria parasites. We also develop a cheap, nondestructive, and high-throughput DNA extraction workflow that provides template DNA from single mosquitoes for the multiplex PCR, which means specimens producing unexpected results can be returned to for morphological examination. Over 1000 individual mosquitoes can be sequenced in a single MiSeq run, and we demonstrate the panel's power to assign species identity using sequencing data for 40 species from Africa, Southeast Asia, and South America. We also show that the approach can be used to resolve geographic population structure within *An. gambiae* and *An. coluzzii* populations, as the population structure determined based on these 62 loci from over 1000 mosquitoes closely mirrors that revealed through whole genome sequencing. The end-to-end approach is quick, inexpensive, robust, and accurate, which makes it a promising technique for very large-scale mosquito genetic surveillance and vector control.

Utility of Complete Mitochondrial Genomes in Phylogenetic Classification of the Species of Anopheles (Culicidae: Anophelinae).

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Among the blood-sucking insects, Anopheles mosquitoes have a very special position, because they transmit parasites of the genus Plasmodium, which cause malaria as one of the main vector-borne disease worldwide. The aim of this review study was to evaluate utility of complete mitochondrial genomes in phylogenetic classification of the species of Anopheles. The complete mitochondrial genome sequences belonging to 28 species of the genus Anopheles (n=32) were downloaded from NCBI. The phylogenetic trees were constructed using the ML, NJ, ME, and Bayesian inference methods. In general, the results of the present survey revealed that the complete mitochondrial genomes act very accurately in recognition of the taxonomic and phylogenetic status of these species and provide a higher level of support than those based on individual or partial mitochondrial genes so that by using them, we can meticulously reconstruct and modify Anopheles classification. Understanding the taxonomic position of Anopheles, can be a very effective step in better planning for controlling these malaria vectors in the world and will improve our knowledge of their evolutionary biology.

Nigeria Anopheles vector database: an overview of 100 years' research.

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Anopheles mosquitoes are important vectors of malaria and lymphatic filariasis (LF), which are major public health diseases in Nigeria. Malaria is caused by infection with a protozoan parasite of the genus Plasmodium and LF by the parasitic worm Wuchereria bancrofti. Updating our knowledge of the Anopheles species is vital in planning and implementing evidence based vector control programs. To present a comprehensive report on the spatial distribution and composition of these vectors, all published data available were collated into a database. Details recorded for each source were the locality, latitude/longitude, time/period of study, species, abundance, sampling/collection methods, morphological and molecular species identification methods, insecticide resistance status, including evidence of the kdr allele, and P. falciparum sporozoite rate and W. bancrofti microfilaria prevalence. This collation resulted in a total of 110 publications, encompassing 484,747 Anopheles mosquitoes in 632 spatially unique descriptions at 142 georeferenced locations being identified across Nigeria from 1900 to

2010. Overall, the highest number of vector species reported included *An. gambiae* complex (65.2%), *An. funestus* complex (17.3%), *An. gambiae* s.s. (6.5%), *An. arabiensis* (5.0%) and *An. funestus* s.s. (2.5%), with the molecular forms *An. gambiae* M and S identified at 120 locations. A variety of sampling/collection and species identification methods were used with an increase in molecular techniques in recent decades. Insecticide resistance to pyrethroids and organochlorines was found in the main *Anopheles* species across 45 locations. Presence of *P. falciparum* and *W. bancrofti* varied between species with the highest sporozoite rates found in *An. gambiae* s.s., *An. funestus* s.s. and *An. moucheti*, and the highest microfilaria prevalence in *An. gambiae* s.l., *An. arabiensis*, and *An. gambiae* s.s. This comprehensive geo-referenced database provides an essential baseline on *Anopheles* vectors and will be an important resource for malaria and LF vector control programmes in Nigeria.

Overwintering of Usutu virus in mosquitoes, The Netherlands.

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Analyses of mosquito-borne virus outbreaks have revealed the presence of similar virus strains over several years. However, it remains unclear how mosquito-borne viruses can persist over winter, when conditions are generally unfavorable for virus circulation. One potential route for virus persistence is via diapausing mosquitoes. We therefore studied whether Usutu virus (USUV), West Nile virus (WNV) and/or Sindbis virus (SINV) can be identified in diapausing mosquitoes in The Netherlands. Mosquito collections were carried out in November 2022 in hibernacula located in two areas with previously observed WNV and/or USUV activity. A total of 4857 mosquitoes, belonging to four species (groups) (*Culex pipiens/torrentium*, *Culiseta annulata*, *Anopheles maculipennis* s.l. and *Culex territans*), were collected. WNV-, USUV- and SINV-screening using a multiplex real-time RT-PCR assay was carried out on mono-specific mosquito pools. One *Culex pipiens/torrentium* pool tested positive for USUV RNA. Whole genome sequencing and subsequent phylogenetic analysis revealed that the virus belongs to USUV lineage Africa 3 and clusters with other USUV sequences derived from The Netherlands in 2022. This finding confirms our hypothesis of the potential of local overwintering of USUV in diapausing mosquitoes in The Netherlands.